

Chapter 1 – Abstract

FraudNet is an AI-powered fraud detection system designed to help merchants detect coordinated fraud rings before financial losses occur.

Unlike traditional fraud detection systems that evaluate transactions individually, FraudNet combines Graph Analytics with Machine Learning.

The project generates a realistic synthetic payment ecosystem, discovers suspicious customer communities, predicts fraud probability using XGBoost, explains every prediction through SHAP, and presents results inside an interactive Streamlit dashboard.

The dashboard provides:

- Real-time payment monitoring
- Community investigation
- Fraud relationship visualization
- Business insights
- AI-powered recommendations
- PDF investigation reports

Chapter 2 – Introduction

Online merchants process thousands of transactions every day.

Most fraud detection systems ask one question:

"Is this transaction suspicious?"

FraudNet asks a different question:

"Is this customer secretly working with other accounts?"

This difference is important because organized fraud usually happens through groups of connected accounts, not isolated transactions.

Instead of analyzing transactions one by one, FraudNet studies relationships between customers.

Chapter 3 – Problem Statement

Merchants face several challenges:

- Refund abuse
- Shared device fraud
- Multiple fake accounts
- Coordinated payment fraud
- False positives

Traditional rule-based systems struggle to identify these hidden relationships.

The project solves this by combining:

- Graph Analytics
- Feature Engineering
- XGBoost
- Explainable AI

Chapter 4 – Objectives

The project aims to:

- Detect coordinated fraud rings.
- Build a customer relationship graph.
- Generate realistic payment data.
- Predict fraud probability using XGBoost.
- Explain predictions using SHAP.
- Help merchants make understandable decisions.
- Visualize suspicious communities.

Chapter 5 – Existing System

Current fraud systems usually rely on:

- Transaction amount
- Location mismatch
- Velocity rules

Limitations

Existing System	FraudNet
Checks one transaction	Checks entire communities
Rule-based	AI-based
Limited explanations	SHAP explanations
No relationship analysis	Graph Analytics
Static monitoring	Live dashboard

Chapter 6 – Proposed System

FraudNet follows a complete AI pipeline.

Complete System Flow

Configuration



Synthetic Dataset Generation



Relationship Graph



Community Detection



Feature Engineering



XGBoost Prediction



SHAP Explainability

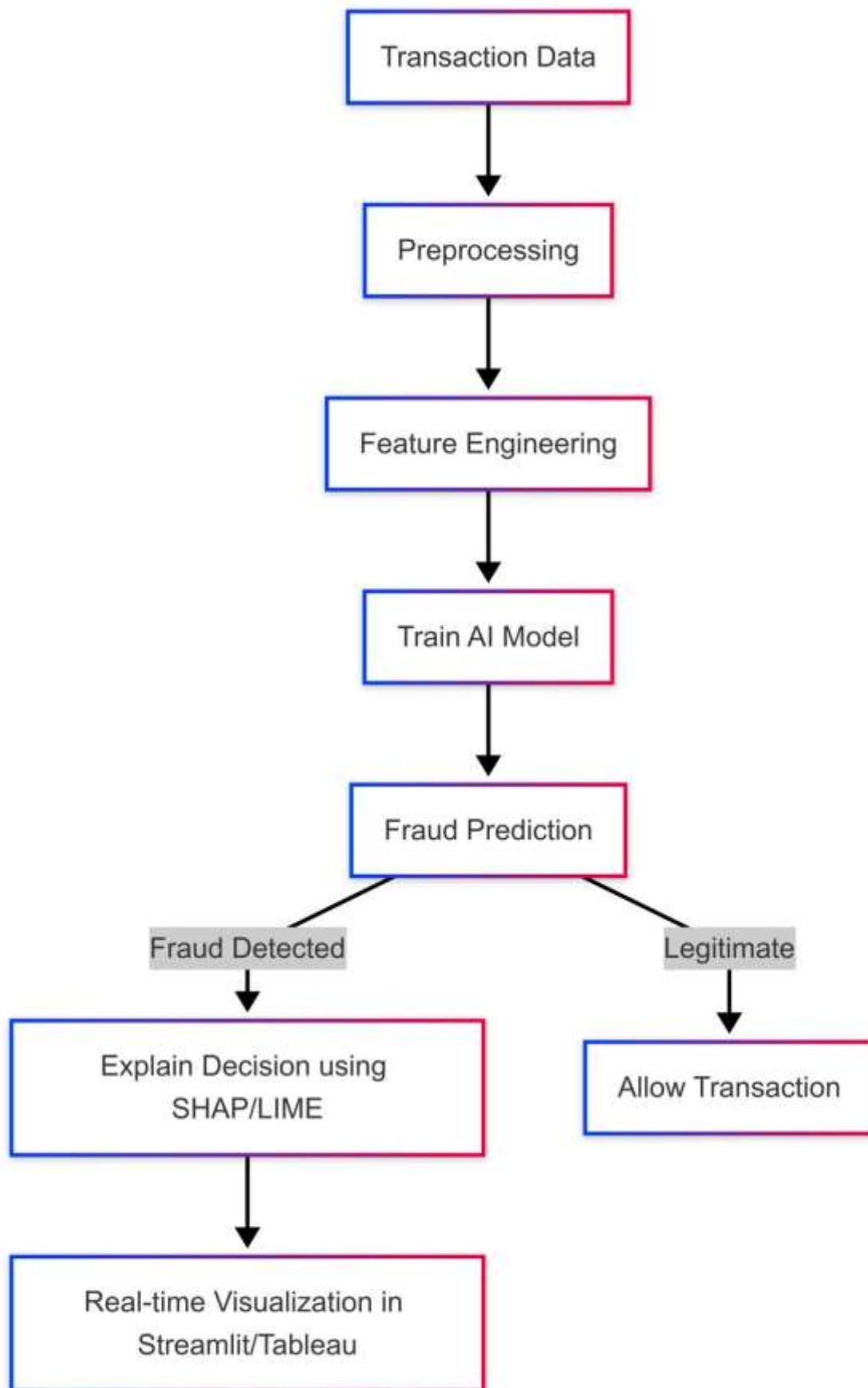


Merchant Dashboard

Chapter 7 – Technology Stack

Component	Technology
Language	Python
Dashboard	Streamlit
Machine Learning	XGBoost
Graph Analytics	NetworkX
Explainability	SHAP
Visualization	Plotly
Data Processing	Pandas
Numerical Computing	NumPy
Reports	ReportLab

Chapter 8 – System Architecture



The system consists of six connected modules.

1. Config
2. Data Generator
3. Graph Detector
4. Train Model
5. Live Simulator
6. Streamlit Dashboard

Each module has one responsibility, making the project easier to maintain.

Chapter 9 – Configuration Module (config.py)

The first file controls the entire project.

Instead of changing numbers across multiple files, every module reads values from one central location. This makes config.py the single source of truth.

FraudNet_Config_Explanation.pdf

Important Parameters

Parameter	Purpose
NUM_ACCOUNTS	1000 customers
NUM_TRANSACTIONS	5000 transactions
NUM_FRAUD_RINGS	8 fraud rings
MIN_RING_SIZE	4 accounts
MAX_RING_SIZE	8 accounts
NORMAL_REFUND_RANGE	2–8% refunds
FRAUD_REFUND_RANGE	15–35% refunds

These values create a realistic payment ecosystem.

Chapter 10 – Data Generation (data_generator.py)

What problem does this file solve?

Real payment data cannot be used because it contains sensitive customer information.

This file creates a realistic synthetic payment ecosystem that includes:

- customer accounts
- transactions

- refunds

The generated data becomes the foundation for the rest of the pipeline.

Generated Files

File	Purpose
accounts.csv	Customer information
transactions.csv	Purchases
refunds.csv	Refund requests

Merchant Database

Five merchants make transactions look realistic:

- TechKart
- StyleHub
- FreshMart
- HomeNest
- FitFuel

Customer Types



The generator creates:

- Normal users
- Households
- Obvious fraud rings
- Moderate fraud rings
- Sneaky fraud rings
- Expert fraud rings

Each behaves differently to mimic real-world behavior.

Meaning of Important Fields

Field	Meaning
account_id	Customer ID
device_id	Customer device
payment_method_fingerprint	Anonymous payment identifier
refund_probability	Likelihood of refund
account_type	Customer category
is_fraud_ring	Ground-truth fraud label

Data Flow

config.py



Generate Accounts



Generate Transactions



Generate Refunds



Save CSV Files



Graph Detector

Chapter 11 – Graph Analytics (graph_detector.py)

What problem does this file solve?

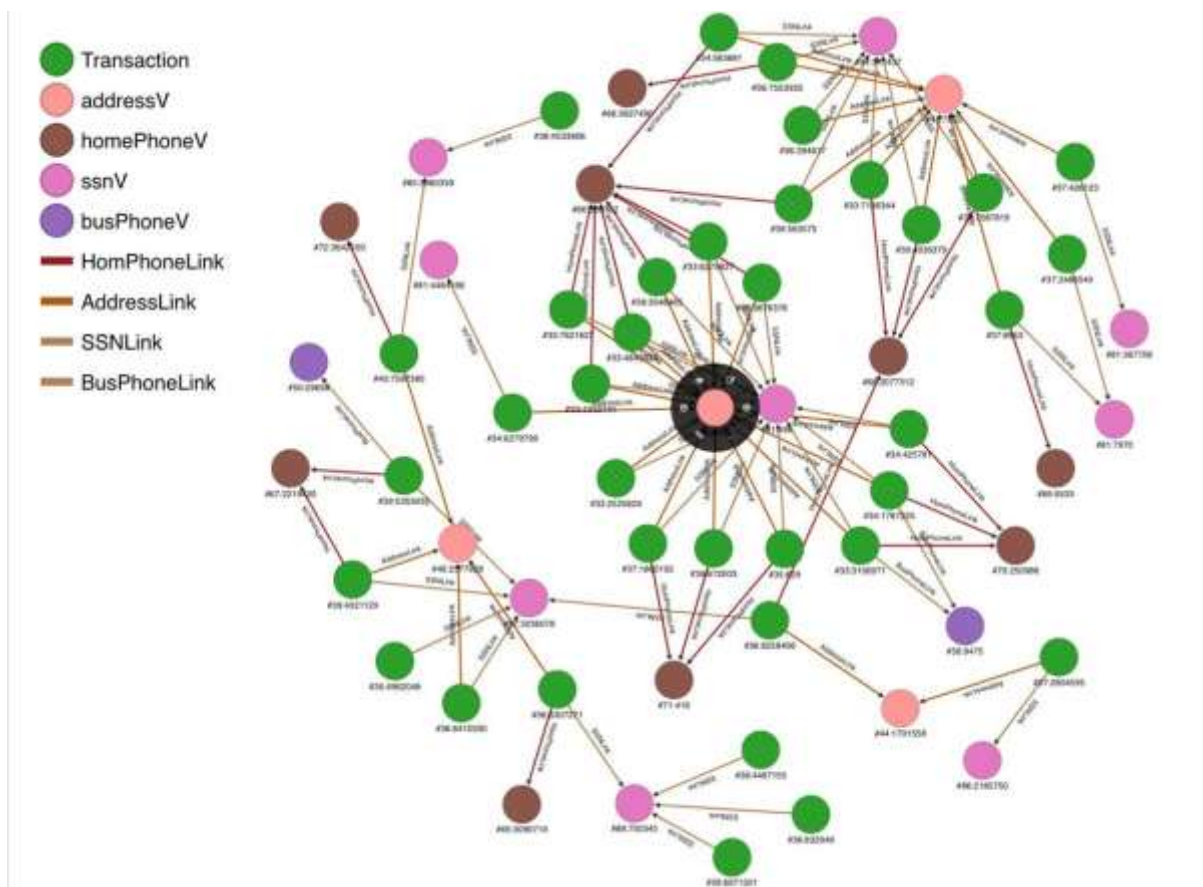
Instead of checking every account independently, FraudNet asks:

"Who is this account connected to?"

Shared devices, payment fingerprints, refund timing, and merchant behavior reveal coordinated fraud rings that row-by-row machine learning can miss.

FraudNet_Graph_Detector_Developer_Guide.pdf

What is a Graph?



- Nodes represent customer accounts.
- Edges represent suspicious relationships.

Relationship Signals

The graph connects customers through:

- Shared devices
- Shared IP addresses
- Payment fingerprints
- Merchant campaigns
- Refund timing

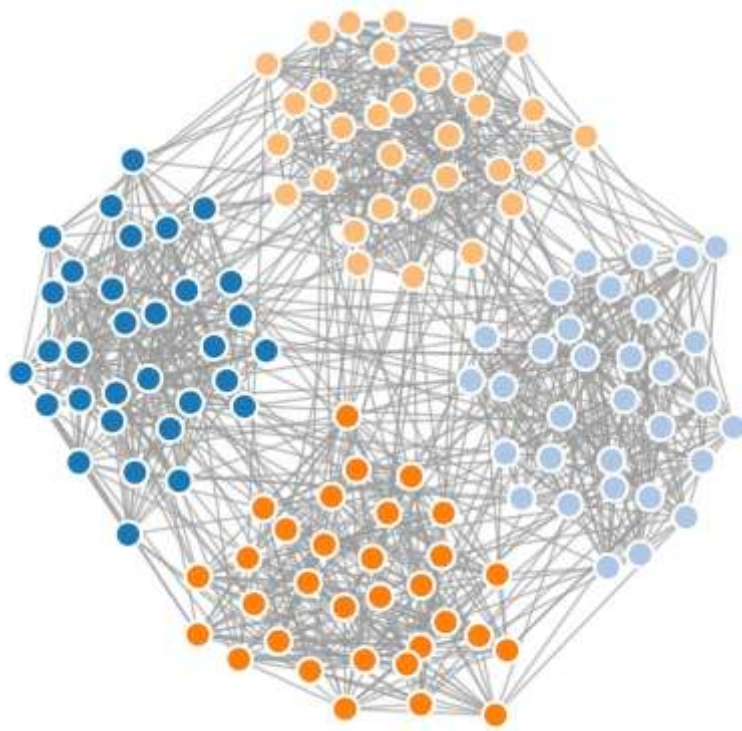
Edge Weights

Signal	Weight
Device	+3
Payment	+3
Refund Timing	+3
IP	+2
Merchant Campaign	+2

Multiple signals strengthen existing relationships.

Community Detection

FraudNet uses the Louvain Algorithm.



Instead of manually searching thousands of customers, the algorithm automatically discovers tightly connected communities.

Community Statistics

For every community the system calculates:

- Size
- Density

- Average account age
- Shared devices
- Shared payment methods

Risk Score Logic

Condition	Points
Community Size ≥ 6	+20
Heavy Device Reuse	+25
Shared Payment Fingerprints	+25
Account Age 0.6	+20

Maximum possible risk becomes 100.

Chapter 12 – Machine Learning (train_model.py)

What problem does this file solve?

The graph detector finds suspicious communities.

However, merchants approve or reject individual accounts, not entire groups.

This file converts graph intelligence into numerical features, trains an XGBoost model, explains predictions using SHAP, and estimates business savings.

ML Pipeline

The complete pipeline is:

CSV Files

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Graph Detection

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Feature Engineering

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Train/Test Split

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XGBoost

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SHAP

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Merchant Cost Simulation

FraudNet_train_model_Complete_Developer_Guide.pdf

Feature Engineering

Machine learning understands numbers, so each account becomes numerical features.

Feature	Why it Matters
Account Age	New accounts may be fraudulent
Community Risk	Brings graph intelligence into ML
Graph Degree	More connections may indicate coordination
Weighted Degree	Stronger suspicious evidence
Refund Rate	Normalizes refund behavior

Train-Test Split

The dataset is divided into:

- 80% Training
- 20% Testing

stratify keeps fraud percentages balanced.

Why XGBoost?

XGBoost builds hundreds of decision trees.

Every new tree learns from previous mistakes, making the model much stronger than a single decision tree.

Important parameters:

Parameter	Value
Trees	200
Max Depth	5
Learning Rate	0.05

Chapter 13 – Explainable AI (SHAP)

Merchants need to know why an account was flagged.

SHAP explains:

- which feature increased fraud probability
- which feature reduced risk

This turns the AI into an explainable system rather than a black box.

Chapter 14 – Model Evaluation

The model is evaluated using:

- Precision
- Recall
- F1 Score
- ROC-AUC

Metric Meaning

Metric	Purpose
Precision	Correct fraud alerts
Recall	Fraud caught
F1	Balance
ROC-AUC	Overall ranking ability

Chapter 15 – Business Threshold Optimization

Instead of maximizing only accuracy, FraudNet estimates business value.

Assumptions:

- Prevented fraud saves ₹5000
- Manual review costs ₹500

Different thresholds are tested to maximize net savings.

Chapter 16 – Live Transaction Simulator

What problem does this file solve?

Historical data makes dashboards look static.

The simulator creates live payment events so merchants can watch:

- Approvals
- Reviews
- Fraud Alerts

appear in real time.

How It Works

1. Select a random account.
2. Choose a random merchant.
3. Generate an amount.
4. Check community risk.
5. Predict fraud.
6. Return a transaction dictionary.

Transaction Fields

Field	Meaning
Time	Current transaction
Merchant	Receiving store
Amount	Payment value
Status	Approved/Review/Fraud
Reason	Human explanation

Chapter 17 – Streamlit Dashboard

The dashboard transforms technical AI outputs into merchant-friendly decisions.

Major sections include:

- Home Dashboard
- Merchant Action Center
- Live AI Payment Monitor
- Business Insights
- Community Deep Dive
- Fraud Relationship Network

- AI Copilot
- PDF Report Export

The merchant sees explanations rather than raw ML numbers.

Chapter 18 – Dashboard Visualizations

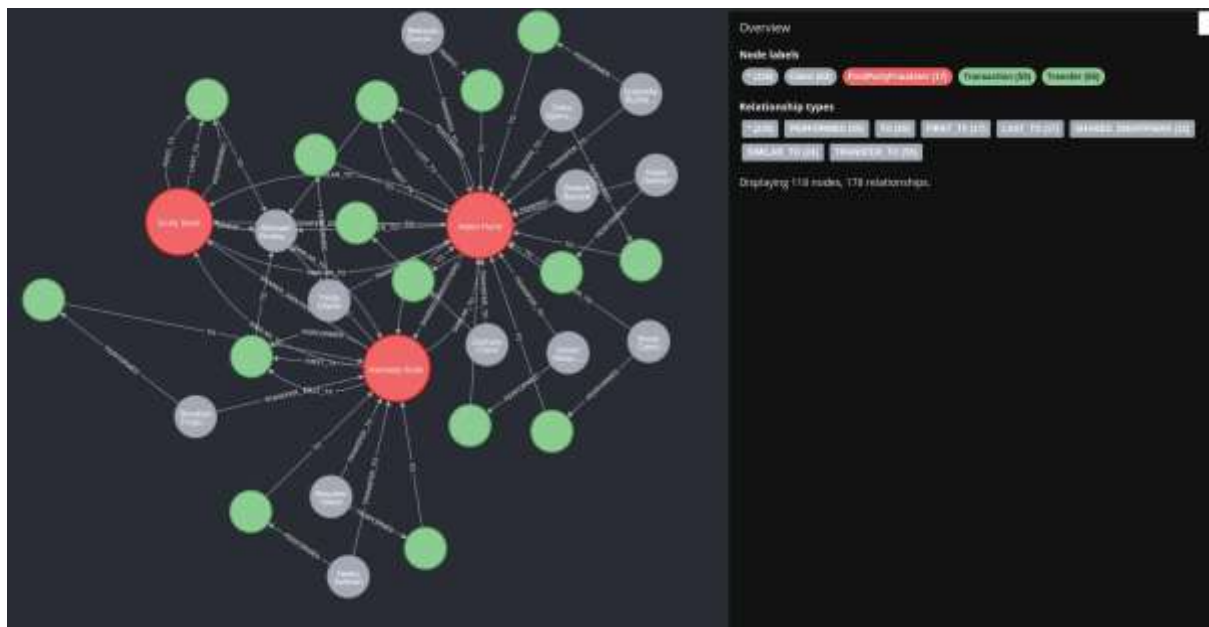
The dashboard displays multiple business insights.

Available Charts

Chart	Purpose
Account Distribution	Customer types
Refund Behavior	Refund patterns
Fraud Probability Distribution	ML predictions
Highest-Risk Communities	Community ranking

The training pipeline also generates ROC, Precision-Recall, Feature Importance, Threshold Savings, and SHAP Summary charts for evaluation.

Chapter 19 – Fraud Relationship Network



The network graph helps merchants understand why accounts were grouped together.

Colors indicate different account types.

Hovering over nodes reveals fraud probability and account information.

Chapter 20 – Business Impact

FraudNet helps merchants by:

- detecting fraud rings earlier
- reducing refund abuse
- prioritizing investigations
- explaining AI decisions
- maximizing business savings

Instead of simply predicting fraud, the system supports real business decision-making through graph intelligence, explainability, and threshold optimization.

Chapter 21 – Future Scope

Future improvements include:

- Real payment gateway integration
- Kafka streaming pipeline
- Graph Neural Networks
- Multi-merchant fraud intelligence
- Mobile merchant dashboard

Chapter 22 – Conclusion

FraudNet demonstrates that fraud detection becomes significantly more effective when Machine Learning and Graph Analytics work together.

The project begins by generating a realistic payment ecosystem, discovers hidden customer communities using weighted relationship graphs, converts graph intelligence into numerical features, predicts fraud using XGBoost, explains every prediction through SHAP, and presents merchant-friendly decisions inside a real-time dashboard.

The combination of synthetic data generation, explainable AI, graph-based community detection, and business-oriented threshold optimization makes FraudNet closer to a practical payment risk engine than a basic fraud classifier